

Marcelo Ellwanger

Software Engineering · QA · Technical Systems

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PROFESSIONAL SUMMARY

Systems-minded software builder with released, field-tested and release-candidate projects across Windows diagnostics, local AI, computer vision, media automation and evidence-driven data systems. Strong in Python, C#/.NET, SQLite, Git/GitHub, debugging, validation and documenting failure modes. Previous electrical and operations work adds hands-on troubleshooting, quality-control and real-world reliability discipline.

TECHNICAL SKILLS

Languages / frameworks: Python · C#/.NET 10 · FastAPI · React · TypeScript · WPF
Data / tooling: SQLite · Git/GitHub · GitHub Actions · Windows diagnostics · REST APIs · local AI / GGUF tooling
Engineering: Automated testing · debugging · telemetry · performance budgets · evidence lineage · reproducible validation · release engineering · documentation

SELECTED ENGINEERING PROJECTS

CrashScope · v0.1 candidate C# / .NET 10 / React / TypeScript / SQLite · Windows diagnostics

- Built a local-first Windows crash-diagnostics platform that correlates rolling CPU/GPU/RAM telemetry with Event Log, WER, watchdog and workload evidence while keeping observation separate from unsupported causation.
- Reached 147 automated tests; real-machine sampler baselines measured 0.0577% background CPU and 0.0578% active CPU, while the dashboard-open Agent measured ~0.068% CPU / ~91 MB working set. CI validates the exact self-contained win-x64 candidate before public beta.

- Designed and released a local-first Windows control deck with GGUF Model Intelligence, safe runtime ownership, hardware telemetry and explainable Smart Launch recommendations.
- Added automated update/repair flows, diagnostics, CI, post-update smoke checks and deployment guards that fail visibly when critical state is invalid.

Retro Game Vision

Computer vision + decision pipeline · Python / SQLite · Field tested

- Built typed data models, provenance-first persistence, compatibility gates, deterministic economics and idempotent validation around messy store-photo/OCR inputs.
- Field-tested at 750+ canonical items, 1,000+ physical observations and 1,800+ market observations while preserving uncertainty and evidence lineage.

EasyFLix · active build

Current pre-1.0 working title: ReelShelf · WPF / .NET 10

- Building a non-destructive Windows media-library application with read-only multi-drive scanning, expandable movie/TV/anime grouping, artwork lookup, source-safe shortcuts and resumable operations.
- Current validation includes 37 passing tests, zero build warnings, dependency auditing, CI and a portable win-x64 artifact; subtitle workflows use exact-episode SubDL selection with private offline Whisper.net fallback and explicit quality gates.

RELEVANT EXPERIENCE

Electrical Trade Experience

Vancouver, Canada · 4,000+ hours

- Hands-on fault finding across electrical/mechanical systems reinforced safe debugging, reproducibility and verification under real-world constraints.

Coffee Manager / Cafe Operations

Alma · Surfers Paradise, Gold Coast, Australia

- Owned scheduling, stock/inventory and specialty-coffee quality control, applying precise process standards in a high-end customer-facing operation.

Customer Service / Teaching

Australia & Japan · selected roles, 2023–2025

- Built communication, training and high-pressure problem-solving skills across hospitality and full-time English teaching.

EDUCATION & CREDENTIALS

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Electrical studies/training - BCIT, Canada · High School Diploma - Queen Elizabeth School, Canada · First Aid & CPR · RSA License

Available for remote software engineering, QA / automation and technical systems roles

